Tarea lección 3.11

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5) a) $\operatorname{sech} 0$

$= \frac{1}{\cosh 0} = 1$

b) $\cosh^{-1} 1$

$= \ln(1 + \sqrt{1^2 - 1}) = 0$

11) $\operatorname{senh}(x+y) = \operatorname{senh}x \cosh y + \cosh x \operatorname{senh} y$

$\operatorname{senh}(x+y) = \operatorname{senh}(x+y)$

15) $\operatorname{senh} 2x = 2 \operatorname{senh} x \cosh x$

$\operatorname{senh}(x+x) = \operatorname{senh}x \cosh x + \operatorname{senh} x \cosh x$

$2 \operatorname{senh} x \cosh x$

23) a) $\lim_{x \to \infty} \tanh x = 1$

b) $\lim_{x \to -\infty} \tanh x = -1$

c) $\lim_{x \to \infty} \operatorname{senh} x = \infty$

d) $\lim_{x \to -\infty} \operatorname{senh} x = -\infty$

e) $\lim_{x \to \infty} \operatorname{sech} x = 0$

f) $\lim_{x \to \infty} \coth x = 1$

g) $\lim_{x \to 0^+} \coth x = \infty$

h) $\lim_{x \to 0^-} \coth x = -\infty$

i) $\lim_{x \to -\infty} \operatorname{csch} x = 0$

j) $\lim_{x \to \infty} \frac{\operatorname{senh} x}{e^x} = \frac{1}{2}$

35) $g(x) = \cosh(\ln x)$

$g'(x) = \operatorname{senh}(\ln x) * \frac{1}{x}$

39) $g(t) = t\coth\sqrt{t^2+1}$

$g'(t) = 1(\coth\sqrt{t^2+1}) - t(\csc^2\sqrt{t^2+1}) * \frac{1}{2\sqrt{t^2+1}} * 2t$

$g'(t) = \coth\sqrt{t^2+1} - \frac{t^2}{\sqrt{t^2+1}}\csc^2\sqrt{t^2+1}$

45) $y = \coth^{-1}(\sec x)$

$y' = \frac{1}{1-(\sec x)^2} * \sec x \tan x$

$y' = \frac{1}{1-(1/\cos x)^2} * \frac{1}{\cos x} * \frac{\operatorname{sen} x}{\cos x} = -\frac{\cos^2 x}{\operatorname{sen}^2 x} * \frac{\operatorname{sen} x}{\cos x} * \frac{1}{\cos x}$

$y' = \frac{-\cos x}{\operatorname{sen}^2 x} * \frac{1}{\cos x} = -\frac{1}{\operatorname{sen} x} = -\csc x$

47) Demuestre que $\frac{d}{dx}\arctan(\tanh x) = \operatorname{sech} 2x$

$= \frac{1}{1+\tanh^2 x} * \frac{1}{\cosh^2 x} = \frac{1}{1+\left(\frac{\operatorname{senh} x}{\cosh x}\right)^2} * \frac{1}{\cosh^2 x} = \frac{1}{\frac{\cosh^2 x + \operatorname{senh}^2 x}{\cosh^2 x}} *$

$\frac{1}{\cosh^2 x} = \frac{\cosh^2 x}{\cosh(2x)} * \frac{1}{\cosh^2 x} = \frac{1}{\cosh 2x}$

$\frac{1}{\cosh 2x} = \operatorname{sech} 2x$